

Lesson 08 - Learning by Recording Cases



Science is built upon facts, as a house is built of stones; but an accumulation of facts is no more a science than a heap of stones is a house.
 – Henry Poincare, *Science and Hypothesis*.

01 - Preview

[Click here to watch the video](#)

Lesson Preview

- Learning by recording cases
- Nearest neighbor method
- Cases in the real world
- k-Nearest Neighbor

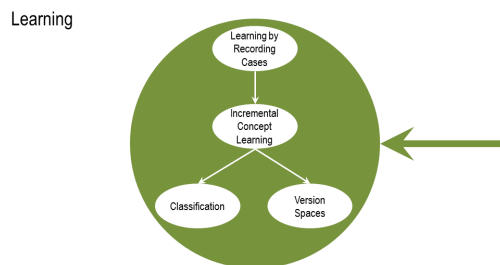


Figure 232: Preview

Figure 234: Preview

Today we'll talk about learning by recording cases. This is our first topic of learning, one of the fundamental elements of knowledge-based AI. It is also a first topic in analogical reasoning, often considered to be a core process of cognition. We'll start talking about recording cases in a general sense. Then we'll discuss a specific method for recording cases called the nearest neighbor method. We'll generalize this method into the k-nearest neighbor method and end by talking about complex cases in the real world.

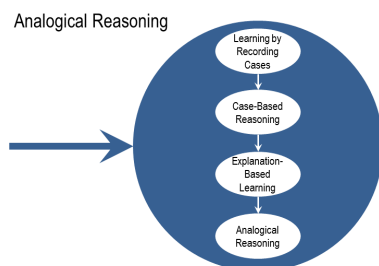


Figure 233: Preview

02 - Exercise Block World I

[Click here to watch the video](#)

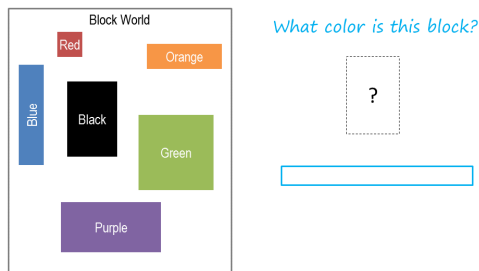


Figure 235: Exercise Block World I

To see how learning, the recording cases might work. Consider a world of blocks. Color blocks, with various shapes and sizes. Six blocks in all. Now let us suppose, that I were to give you a question. So, based on your experiences in this world. What do you think is the color, of this block?

03 - Exercise Block World I

[Click here to watch the video](#)

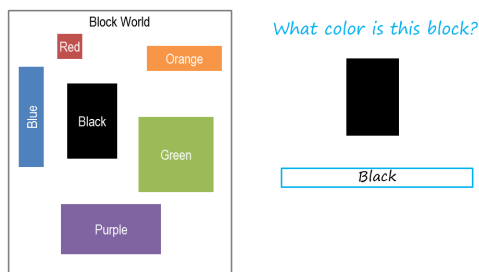


Figure 236: Exercise Block World I

You're right, David. Of the various blocks given here, this block best resembles the black block. And therefore the best guess would be, this is in fact, a black block. This is an example of learning by recording cases, because six cases were recorded in the agent's memory. So when now when a new problem comes along, then the agent gives an answer to that new problem based on the cases that it already had recorded in its memory, it simply sees which case most closely resembles the new situation. And gives the answer to the new situation for that most closely resembling case.

04 - Learning by Recording Cases

[Click here to watch the video](#)

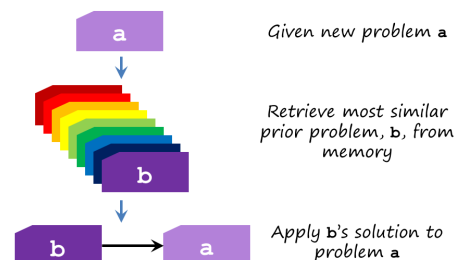


Figure 237: Learning by Recording Cases

This gives a visualization of what is happening in learning by recording cases. There's a new problem *a* shown by the color here, and memory contains a large number of cases, again represented by the different colors here. So we retrieve the case *b* that is most similar to the new problem *a*. In this particular case, we're deciding by color. And whatever was the solution to *b*, we apply to the situation *a*. While this visualization is useful, it's still very abstract. Let's think in terms of some practical, everyday examples. So you get up in the morning and you want to go for a run. You put on your sneakers, you have to tie your shoelaces. Well, how do you tie your shoelaces? That's a new problem. But of course, you have tied shoelaces many, many times earlier. So you have a memory of tying shoelaces to different kind of shoes. That's all's to it is cases in your memory. So as you start tying the shoelaces for the shoe today, you simply retrieve the closest matching case and apply it. None of us really thinks very hard in the morning about how exactly to tie shoelaces. If you were to do it, it would take us a very long time. So in learning by recording cases, memory guardedly supplies us with the answer. We don't have to think about it. So another example of this that comes to mind is in programming. Oftentimes in programming, we deal with the same type of problem over and over again. So I might imagine I'm starting out a new program and I'm writing a program in Java. So I'm given a new problem of creating a new program in Java. All I'm going to do is look back in my memory of cases and retrieve another case of starting a new program in Java. And I'm going to apply the solution to that program directly to this one. In this case,

it's going to be something like, when I'm starting a new program in Java, last time, I started with public static void main. I'm going to apply that solution directly to my new problem, and it works. The solution can just be transferred directly to the new problem without any kind of modification. Then later in developing that same program, I might hit, for example, a null pointer error. I'm going to use that null pointer error as a new problem and use it as a probe into my memory. I'm then going to retrieve a case of when I encountered a null pointer error in the most similar program I've worked on, and then that's going to give me a solution that I potentially can apply directly to my current problem. So for example, the solution to the last time I encountered that same error might be to run the program in debug mode and allow it to tell me exactly what variable is null at the time of execution. That's a good example, David. And we could even try to generalize it to medical diagnosis. Imagine that you went to a medical doctor with a set of signs and symptoms. So the doctor is faced with a new problem. What is a diagnosis for your signs and symptoms? The doctor may even have a number of cases recorded in her memory. These are the cases she has encountered during her experience. So the doctor must select the most similar case, the most closely resembling case, which in this case might be b, and say that I will apply to a exactly the same diagnosis that I applied to b. So a case then is an encapsulation of a past experience. And learning by recording cases is a very powerful method that works in a very large number of situations ranging from tying your shoelaces to medical diagnosis.

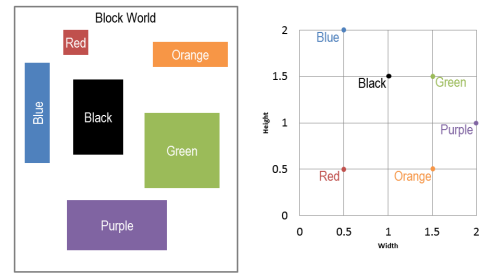


Figure 238: Case Retrieval by Nearest Neighbor

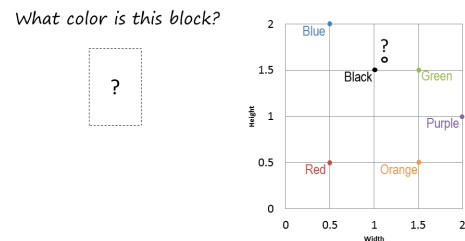


Figure 239: Case Retrieval by Nearest Neighbor

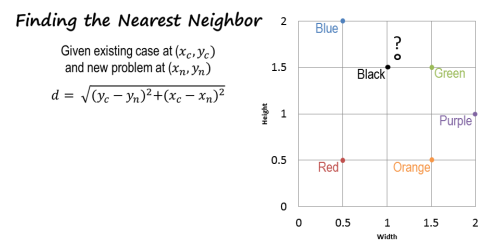


Figure 240: Case Retrieval by Nearest Neighbor

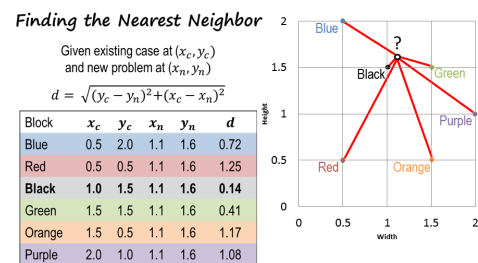


Figure 241: Case Retrieval by Nearest Neighbor

05 - Case Retrieval by Nearest Neighbor
Click here to watch the video

Let us look at this learning by the recalling cases a little bit more closely. Implicit in our discussion so far has been the notion of most

similar or most closely resembling. But how can we operationalize it? How can we make it more explicit? So once again here is a world of various colored blocks and we can represent these various blocks back to the notion of knowledge for presentation. We can represent our knowledge of these various blocks. In a two dimensional grid, the width of the block and the height of the block. So the blue block may right here, the red block here and so on. So when the new problem comes along we may represent it on the same two dimensional grid. In this particular case the new problem might have been represented in this particular dot. Now given all the cases in the new problem we may calculate the distance between the new problem to each of the previous known cases. Once we have calculated between the new problem and each of the previous cases we can simply select a case which is closest to the new problem. This method is called the nearest neighbor method. Now we need a way of calculating the distance between a problem, in that case. One measure of the distance is called the Euclidean distance. Here is a formula for the Euclidean distance. The Euclidean distance between two points, x of c and y of c, which define the case, and the x of n and the y of n, which define the problem, is given by this formula. Now we can easily calculate the Euclidean distance between each of the cases and new problem, and this table summarizes the distances. Given this table, we can very quickly see that the case of the black block is closest to the new problem and therefore one might give the answer the new block is also black in color. So the nearest neighbor method is one method of finding the most similar case or the most closely resembling case.

06 - Exercise Retrieval by Nearest Neighbor

Click here to watch the video

What color is this block?

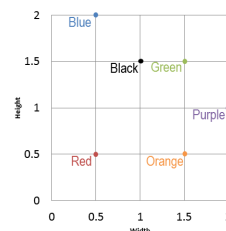
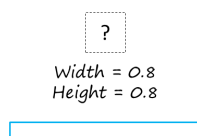


Figure 242: Exercise Retrieval by Nearest Neighbor

Let us do an exercise together. Given the block shown here with the width of 0.8 and the height of 0.8, what do you think is the color of this block?

07 - Exercise Retrieval by Nearest Neighbor

Click here to watch the video

What color is this block?

Block	x_c	y_c	x_n	y_n	d
Blue	0.5	2.0	0.8	0.8	1.24
Red	0.5	0.5	0.8	0.8	0.42
Black	1.0	1.5	0.8	0.8	0.72
Green	1.5	1.5	0.8	0.8	0.98
Orange	1.5	0.5	0.8	0.8	0.76
Purple	2.0	1.0	0.8	0.8	1.22

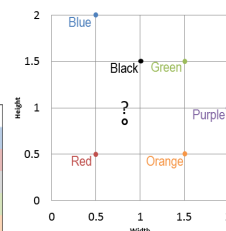


Figure 243: Exercise Retrieval by Nearest Neighbor

Finding the Nearest Neighbor

Given existing case at (x_c, y_c)
and new problem at (x_n, y_n)

$$d = \sqrt{(y_c - y_n)^2 + (x_c - x_n)^2}$$

Block	x_c	y_c	x_n	y_n	d
Blue	0.5	2.0	1.1	1.6	0.72
Red	0.5	0.5	1.1	1.6	1.25
Black	1.0	1.5	1.1	1.6	0.14
Green	1.5	1.5	1.1	1.6	0.41
Orange	1.5	0.5	1.1	1.6	1.17
Purple	2.0	1.0	1.1	1.6	1.08

Block	x_c	y_c	x_n	y_n	d
Blue	0.5	2.0	1.1	1.6	0.72
Red	0.5	0.5	1.1	1.6	1.25
Black	1.0	1.5	1.1	1.6	0.14
Green	1.5	1.5	1.1	1.6	0.41
Orange	1.5	0.5	1.1	1.6	1.17
Purple	2.0	1.0	1.1	1.6	1.08
Blue	0.5	2.0	0.8	0.8	1.24
Red	0.5	0.5	0.8	0.8	0.42
Black	1.0	1.5	0.8	0.8	0.72
Green	1.5	1.5	0.8	0.8	0.98
Orange	1.5	0.5	0.8	0.8	0.76
Purple	2.0	1.0	0.8	0.8	1.22

Figure 244: Exercise Retrieval by Nearest Neighbor

So in this problem, we are dealing with a two-dimensional grid, because here, two coordinates, x and y, are enough to represent any one point. In the real world, of course, problems are not

that easy to represent, and one might need a multi-dimensional space in order to be able to represent all the cases in the new problem. Let's examine a problem like that now.

08 - Exercise Recording Cases in Real Life

[Click here to watch the video](#)



Figure 245: Exercise Recording Cases in Real Life

So, here is a map of a small portion of Long Island, New York. Imagine there is an automated car that can navigate the streets of this neighborhood. It comes from the factory with these six cases bootstrapped in it. A, B, C, and so on. For the time being, assume that the car navigates its way in this neighborhood solely by the method of learning where the car in case is. So all it can use is this cases that it knows about. Now, suppose that we have a new problem. The new problem is how to go from Q to this end destination denoted by the arrow. What route is most similar to this new problem?

09 - Exercise Recording Cases in Real Life

[Click here to watch the video](#)



Figure 246: Exercise Recording Cases in Real Life

What do you think, David? So personally I said that D is the most similar case. There are a couple of cases that are closer to the origin, like A and B, but they both go in a very different direction than we're trying to go with Q. Similarly, E ends even closer to Q, but E starts much further away. D looks like it's the one that both starts closest to Q and ends closest to Q. D here is the right answer. Let us think how we can program an air agent to come up with this answer.

10 - Nearest Neighbor for Complex Problems

[Click here to watch the video](#)

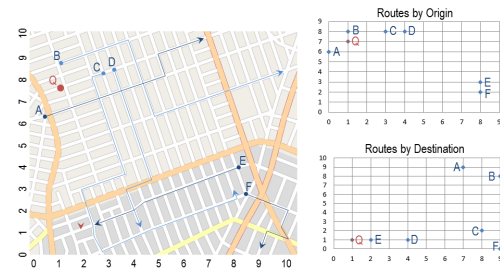


Figure 247: Nearest Neighbor for Complex Problems

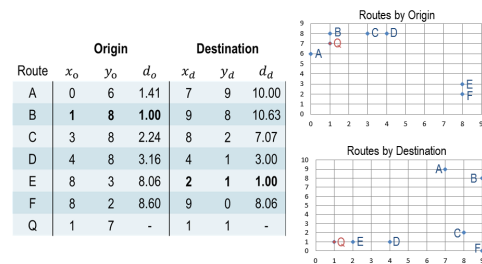


Figure 248: Nearest Neighbor for Complex Problems

Now we can try to calculate the most similar case of the new problem based solely on the origin. The two dimensional grid here tries to represent both all the cases and in your problem. Of course we can also calculate the similarity of the new problem with the old cases based on destination. This two dimensional grid captures the cases and the problem based on the destination. You can compute the [Euclidean] distance from Q in all the cases placed on the origin, shown

here. And you can do the same thing with the destination, shown here. If we focus only on the origin, then the B case seems the closest. If we focus solely on the destination, the E case seems the closest. However, the B case is not very good when we look at the destination. And the E case is not very good when you look at the origin. How then might an AI agent find out which is the best route of all of these choices? How might it decide D is the best route?

11 - Nearest Neighbor in k-Dimensional Space

[Click here to watch the video](#)

Finding the Nearest Neighbor

Given existing case at (x_c, y_c)
and new problem at (x_n, y_n)

$$d = \sqrt{(y_c - y_n)^2 + (x_c - x_n)^2}$$

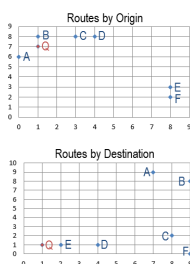


Figure 249: Nearest Neighbor in k-Dimensional Space

Finding the Nearest Neighbor

Given existing case at (x_c, y_c)
and new problem at (x_n, y_n)

$$d = \sqrt{(y_c - y_n)^2 + (x_c - x_n)^2}$$

Given existing case at $(c_1, c_2 \dots c_k)$
and new problem at $(p_1, p_2 \dots p_k)$

$$d = \sqrt{\sum_{i=1}^k (c_i - p_i)^2}$$

Route	Origin			Destination		
	x_o	y_o	d_o	x_d	y_d	d_d
A	0	6	1.41	7	9	10.00
B	1	8	1.00	9	8	10.63
C	3	8	2.24	8	2	7.07
D	4	8	3.16	4	1	3.00
E	8	3	8.06	2	1	1.00
F	8	2	8.60	9	0	8.06
Q	1	7	-	1	1	-

Figure 250: Nearest Neighbor in k-Dimensional Space

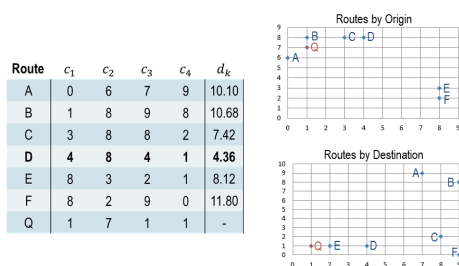


Figure 251: Nearest Neighbor in k-Dimensional Space

Route	c_1	c_2	c_3	c_4	d_k
A	0	6	7	9	10.10
B	1	8	9	8	10.68
C	3	8	8	2	7.42
D	4	8	4	1	4.36
E	8	3	2	1	8.12
F	8	2	9	0	11.80
Q	1	7	1	1	-

Route	Origin			Destination		
	x_o	y_o	d_o	x_d	y_d	d_d
A	0	6	1.41	7	9	10.00
B	1	8	1.00	9	8	10.63
C	3	8	2.24	8	2	7.07
D	4	8	3.16	4	1	3.00
E	8	3	8.06	2	1	1.00
F	8	2	8.60	9	0	8.06
Q	1	7	-	1	1	-

Figure 252: Nearest Neighbor in k-Dimensional Space

Earlier we had this formula for calculating the Euclidean distance in two dimensions. Now we can generalize it to many dimensions. So here is a generalization of the previous formula computing nearest neighbor. In this new formula, both the case and the problem are defined in K dimensions. And we'll find the Euclidean distance between them in this K space. So this table summarizes Euclidean distance between the cases and the new problem in this multidimensional space where we are dealing both with the origin and the destination, and where the origin as well as the destination are specified by the x and y coordinates. Looking at this table, we can very quickly see that D and not B or E, is the closest case, your most similar case, linear problem Q. This method is called the KNN method where NN stands here for nearest neighbor, K nearest neighbor method. This is a probably method as simple as it is. Of course, it also has limitations. One limitations is that, in the real world, the number of dimensions in which I might want to compute the distance between the new problem and old cases might be very large, a high dimensional low space. In such a situation, deciding which of the stored cases is closest to the new problem may not be as simple as it appears here. A second difficulty with this method is, that even if the new problem isn't very close to an existing case, that does not mean that the existing cases solution can or should be darkly applied to the new problem. So, we need both alternative methods of retrieving cases from memory, and methods for adapting passed cases to fit the requirements of the new problem. That

is called [Case-Based Reasoning] and we will discuss that in the next lesson.

12 - Assignment Learning by Recording Cases

[Click here to watch the video](#)

Assignment

How would you use recording cases to design an agent that could answer Raven's Progressive Matrices?

Figure 253: Assignment Learning by Recording Cases

For this assignment, talk about how you might use notion of recording cases to design an agent that can solve Raven's Progressive Matrices. You might think of cases in a variety of different ways here. For example, each figure in a problem could be a case. Each transformation between figures could be a case. Or more broadly, each problem that your agent has encountered in the past could be a case. As part of this, you'll also need to think about how to evaluate similarity. If you're using figures, how do you evaluate the similarity between two figures in a problem? Or how do you evaluate the similarity between two transformations and a problem? Or more broadly, how do you find what problem that you face in the past, is most similar to the new one you're facing now?

13 - Wrap Up

[Click here to watch the video](#)

To recap...

- Recording and using cases
- Nearest neighbor method
- Cases in real-world problems
- Nearest neighbor in k -dimensional problems

Figure 254: Wrap Up

So today we discussed a learning method called learning by recording cases. In learning by recording cases, we file away individual cases we have encountered in the past in order to use them for future problem solving. We talked about the nearest neighbor method as a way of finding the most similar case to the current problem that we faced in the past. But in the real world, this can often be very difficult. So we talked about using nearest neighbor to find very complex similar cases to our current problem, such as our navigation example. However, there are still a lot of limitations to this method. Oftentimes, just executing a solution we've used in the past doesn't work. And oftentimes, we have to store cases based on qualitative labels instead of numeric labels. These weaknesses will be addressed in our next lesson when we talk about case-based reasoning. There we'll add adaptation and evaluation into our process, and start to be able to use cases in a much more thorough and robust way.

14 - The Cognitive Connection

[Click here to watch the video](#)

Learning by storing cases in memory has a very strong connection to cognition. Cognitive agents like you and I are situated in a world. Our interactions with the world have certain patterns of regularity. The world offers us the same problems again and again. If we think about it, the kinds of problems that you and I deal within a routine everyday basis are the same problems that occurred yesterday and the day before. Tying shoelaces is a good example of that. When we have to tie shoelaces, none of us thinks a lot about how to do it. Memory supplies us with the answer. We don't think as much as we think we do. If you recall we have drawn a cognitive architecture earlier that had three components in it, reasoning, memory, and learning. When we think of intelligence, we typically focus on the reasoning component. We think intelligence has to do with reasoning, with solving problems, with decision making. To some degree, that is true. By learning by recording cases, shifts the balance between the component. It says that, learning is very important and so is memory. We

recall things in memory and then memory supplies us with the answers so that we don't actually have to reason as much as we think we need to.

15 - Final Quiz

[Click here to watch the video](#)

Please write down what all you learned in this lesson, in this box.

16 - Final Quiz

[Click here to watch the video](#)

And thank you for doing it.

Summary

This lesson cover the following topics:

1. Memory is as important as Learning/Reasoning so that we can fetch the answer to similar cases encountered in the past and avoid having to redo the non-trivial task of learning and reasoning, thereby saving effort.
2. A case is an encapsulation of a past experience that can be applied to a large number of similar situations in future. The similarity metric can be as simple as the Euclidean distance metric or a complex metric involving higher dimensions.
3. kNN method is one method to find the most similar case from memory for a new problem.

4. In some cases, we need to adapt the cases from our memory to fit the requirements of the new problem. In some cases, we also need to store cases based on qualitative labels along with numeric labels to make the comparison applicable for particular situations.

5. Learning by storing cases in memory has a very strong connection to Cognition since human cognition works in a similar manner by recording cases and applying them to new problems in real world by exploiting the patterns of regularity in them.

References

1. Winston P., Artificial Intelligence, Chapter 19.

Optional Reading:

1. Winston Chapter 19; [Click here](#)
2. These Robots Learn How to Cook by Watching YouTube; [Click here](#)
3. The wonderful and terrifying implications of computers that can learn; [Click here](#)

Exercises

None.